Design of a Multi-purpose Surface-EMG Readout System for Draft Control Applications

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***Abstract*—This paper presents current work on the design of a multi-purpose surface-electromyography (EMG) readout system that can be used in various draft controlled applications of low-accuracy (e.g. rehabilitation environments for stroke patients, ambient assisted living for elders, remote control of robotic arm). The proposed system interfaces six low-cost and low-power EMG channels exhibiting ±1.5mV sensitivity and 400mV output range. The ADC converters of a microcontroller (i.e. Arduino) platform are used to fetch the analog signal from the read-out system. Digital data is verified after ADC conversion using oscilloscope instruments. The microcontroller is programmed to filter the analog signal and identify its rough changes based on continuously adjustable thresholds. The measured data are transmitted in a flexible digital coded format via the microcontroller's network connectivity hardware to the associated control receiver, which in turn decodes it and performs the associated operation as required by the draft-controlled environment. The current consumption ranges from 330uA to 2.9mA per channel, depending on the type of operational amplifier used.**

Keywords—multi-channel surface-EMG, read-out circuits, draft control applications, opamp staging, microcontroller

I. Introduction

Modern draft control applications inherently maintain a cyber-physical nature [1], which is often supported by the incorporation of smart wearable devices. A specific sector of such systems employs the usage of EMG sensors that are consequently interfaced to data processing units via read-out circuits. Within this context, various EMG based environments have been proposed. Authors in [2] presented a control technique with high accuracy to achieve anthropomorphic functions in assistive robot arm application using an EMG-based classification method. Similarly, researchers in [3] proposed a method in which the EMG signals are recorded using surface EMG electrodes placed on the human's skin while real-time experiments, including random arm motions in the 3-D space with variable hand speed profiles are incorporated to capture EMG changes over time.

In addition to the previously mentioned robotic arm applications, the work in [4] exploits the combination of a 5-channel surface EMG and a 3-D accelerometer towards the implementation of bed-set for the automated Greek sign language gesture recognition in deaf individuals. Furthermore, EMG based environments have been used autonomously in patients to monitor their rehabilitation-physiotherapy courses [3, 5, 6, 7] or for the detection of sleep bruxism activity [8]. Moreover, intraoperative neuro-monitoring method for the diagnosis of neurological disorders assisted by EMG environments was presented in [9]. The comprehension of the motor neuron signals was studied in an EMG data recording system [10] while a similar approach facilitating an 8-channel EMG system for noninvasive usage has also been presented in [11].

Drawing from the above context, current work aims to present the design of a low-cost and low-power surface-EMG readout device that supports 6-EMG channels. In comparison to previous designs, our approach targets to capture rough changes of the signal rather than focusing in taking high precision measurements, since such (rough) changes suffice for draft controlled environments and applications addressed in this work. As soon as these changes are sensed and identified, the system is programmed to send a specific configuration dictionary to the draft control receiver that could handle for example the rehabilitation of patients (e.g. upper limb for stroke patients), the regulation of the ambient environment for elder people, robotic arm control applications etc. The benefit of the proposed approach is the reduction of the number of stages per EMG channel since just the output of an instrumentation amplifier is required for detecting the (rough) signal changes needed for the targeted applications. The necessary operations of the subsequent stages of an ordinary EMG readout circuit are performed in software by the employed microcontroller unit, lowering the cost and the power consumption. The rest of the paper is organized as follows: Section 2 describes and analyzes the structure of the read-out circuit. The architecture of the system and its configuration modes are given in section 3 while section 4 provides future directions and concludes the article.

II. EMG Readout Circuitry

An EMG measurement is usually taken by sensing the conduction of the electrical potential between two skin electrodes that are connected to a muscle. To this end, a specific electronic device (read-out circuit) is required to acquire the analog signal, amplify and properly condition it, and eventually feed it to an Analog to Digital Converter (ADC) usually forming the front-end of a microcontroller platform for further processing in digital domain which in our case is an Arduino based one. Since our target is to identify rough EMG signal fluctuations, we have been differentiated from the conventional read-out circuitry [3, 4, 5, 6] by focusing on decreasing the number of opamp stages towards the direction of decreasing power-consumption and lowering the implementation cost. Fig. 1 shows the stages of an ordinary EMG read-out circuitry.

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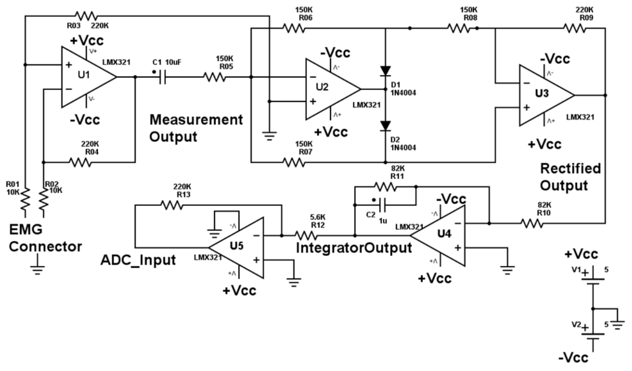

Fig. 1. The stages of an ordinary EMG readout circuit.

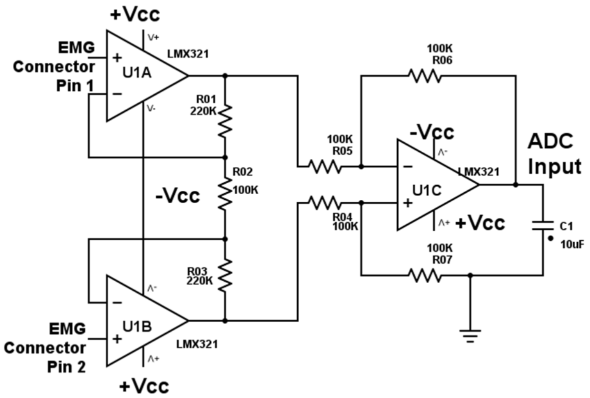

Fig. 2. Instrumentation amplifier used as a first stage of the EMG readout circuit.

The read-out circuit connects to the EMG electrodes and follows the amplitude of the signal which relates to the muscle contraction level. The electrical difference between the two electrodes is amplified in order to attain a discernible difference. The first measurement stage amplifies the input voltage difference and should exhibit high input impedance to couple to body's relatively high impedance. A simple architecture for the first stage that achieves this function is shown in Fig.1, where only one operation amplifier is used. Unfortunately, such a voltage difference amplifier (Fig. 1) does not have adequately high input impedance and generates an oscillation caused by the charge of the user skin. For this reason, the architecture of the so called instrumentation amplifier shown in Fig. 2 is usually employed as the voltage difference amplifier in the first stage of the majority of read-out implementations for EMG applications and this also holds for our case. Indeed, the instrumentation amplifier like the one shown in Fig. 2 is more appropriate since its input impedance is theoretically infinite. The output of this stage is a positive of negative offset disturbed by the signals of the sensor's skin surface electrodes. Capacitor C1 removes any positive or negative Direct Current (DC) offset and allows an AC signal to be rectified by the following two opamps of Fig. 1 (U2 and U3). The integrator U4 performs the envelope detection of the single ended signal. The last stage (U5) amplifies the output of U4 bringing it to an appropriate range for the subsequent ADC. In this work we will explain why the first stage (implemented as shown in Fig.2) is the only necessary one for the targeted applications.

Fig. 3. Attachment of the EMG electrodes on the arm.

If R01=R03=R, R02=R_{gain} and R04=R05=R06=R07 in the instrumentation amplifier of Fig. 2 then the gain A_v of this stage is:

$$A_v = 1 + \frac{2R}{R_{gain}} \tag{1}$$

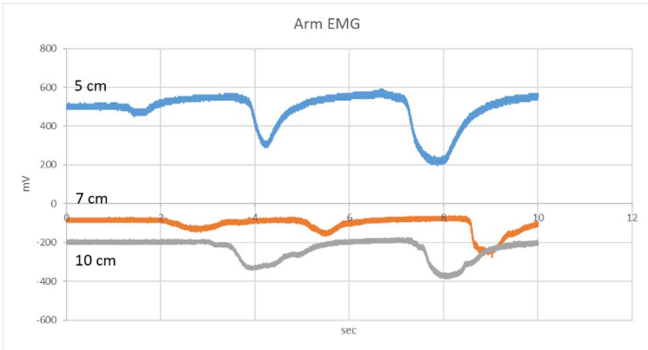

Fig. 4. Output of the instrumentation amplifier of Fig. 2 for three different electrodes' distances (5 cm, 7cm and 10 cm).

The EMG sensor has been attached to the arm muscle as shown in Fig. 3. The ground is the black electrode shown at the bottom. The position of the other two electrodes and their distance, determines the DC offset as shown in Fig. 4. Three distances have been tested in Fig. 4: 5 cm, 7 cm and 10 cm. In all these three cases the muscle was contracted with three different levels of strength (low, medium, high). The contraction with high strength generated a signal amplitude of up to 400 mV using a ±5 V power supply. In this range of 400 mV two or three draft levels of contraction can be safely recognized at the output of an ADC that accepts such a signal at its input provided that the DC offset is known. We choose to estimate this offset in real time as will be explained later. As can be seen from Fig. 4, the DC offsets are approximately 500 mV, -80mV and -200 mV for 5 cm, 7 cm and 10 cm electrodes' distance respectively. This means that the overall signal range can be kept either within positive or negative values if the distance between the electrodes is selected appropriately. For example, the electrodes' distance can be selected somewhere between 3 cm and 6 cm to ensure that the signal will have only positive values. The DC offset will depend on the exact electrodes' distance but no negative values will appear.

Fig. 5. The test setup of the EMG readout circuit shown in Fig. 2 (without C1).

As aforementioned the output signal of Fig. 2 (the differentially amplified EMG signal) feeds an ADC channel of an Arduino Zero based platform. The 6 ADC channels of Arduino Zero have a configurable resolution of up to 12-bits which is much higher than the resolution required by the targeted applications. The typical voltage supply of the Arduino module is 5 V matching one of the supplies of the EMG readout circuitry. However, the actual voltage supply of the Arduino processor after regulation is 3.3 V and thus, the ADC input should be preserved in the range [0, 3.3 V]. Provided that the offset is preserved within this range using acceptable electrodes' distances and it is detected by the software of the Arduino controller, there is no need to use a capacitor like C1 of Fig. 1 to convert the measured signal to AC and then rectify/amplify it with U2-U5 opamps. The achieved signal amplitude of 400 mV may not take full advantage of the 3.3 V input range of the Arduino ADCs but this is not an issue for the resolution required by the targeted draft control applications. The extended 3.3 V input range of the Arduino ADC could be exploited by allowing the 400 mV output range of the EMG readout circuit of Fig. 2 to be shifted up or down within this range. This fact allows the system to be tolerant to the selected electrodes' distance avoiding the need to bring first the signal around 0V (using C1 of Fig. 1) and then rectifying the signal using U2 and U3 as shown in Fig. 1. The integrator U4 of this figure is also redundant since a moving average performed by the software of the Arduino microcontroller can track the envelope of EMG signal. The inverting amplifier of U5 of Fig. 1 is also avoided since the 400 mV amplitude provided by the instrumentation amplifier of Fig. 2 is adequate to safely detect 2 or 3 EMG signal levels using appropriate (adjustable in real time) thresholds.

Implementing the circuit of Fig. 1 using classic UA741 operational amplifiers leads to a current consumption of 14 mA that can be reduced to 750 uA if a low power LM321 operational amplifier is used. This current consumption is extended to 19.6mA (or 1.05mA) if the first stage of Fig. 1 is replaced by the instrumentation amplifier of Fig. 2 and UA741 (LM321) are employed. Using a single instrumentation amplifier per EMG channel as described earlier and shown in Fig.6, reduces the current consumption to a range between 320uA and 450uA if three LM321 opamps or a single ADA4692 instrumentation amplifier is used. The test setup of a single EMG channel implemented on a breadboard as the circuit of Fig. 2 is shown in Fig. 5. It is interesting to note in this figure that the waveform measured by the oscilloscope and displayed in the laptop screen has severe glitches. These glitches are suppressed when the decoupling capacitor C1 of Fig. 2 is used as shown by the curves of Fig. 4.

III. Architecture and Functionality of the EMG Based System

The read-out circuit is the key component in the proposed surface-EMG based system. In a multi-channel EMG acquisition environment, multiple pairs of EMG sensor and read-out devices are interfaced to the ADC converters used in microcontroller (uC) platforms. Fig. 6 presents a 6-channel surface-EMG prototype which obtains the body measurements, transforms them into a digital datum and transmits it via network connection devices to the associated draft-control unit.

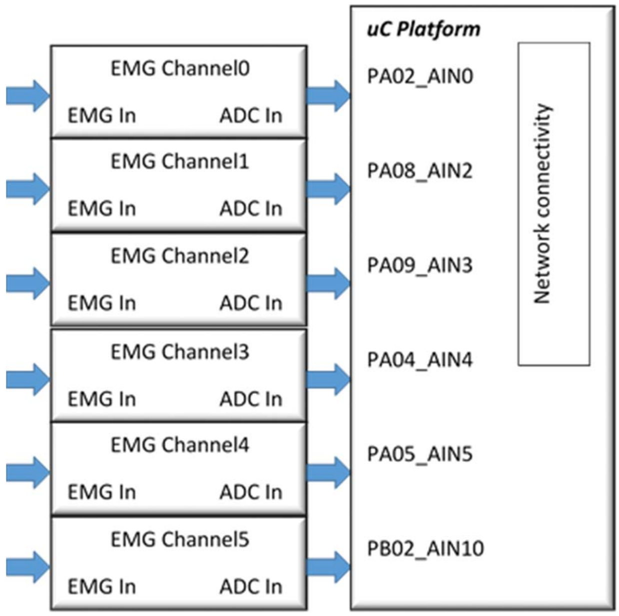

Fig. 6. The block-diagram of the multichannel EMG Controller.

The proposed multichannel EMG architecture shown in Fig. 6, requires 6 banks of read-out circuits supported by equivalent number of Symmetric power supply devices. Each one of this bank connects to the analog ports *AIN* of the microcontroller platform (Fig. 6 illustrates Arduino Zero platform I/Os [13]). As already mentioned, Arduino Zero boards support ADCs with 0-3.3V input range (Arduino Uno supports 0-5V input range) and up to 12-bit resolution. Although the maximum clock frequency that Arduino Zero supports is 48MHz, we have configured it to operate at 16MHz nominal clock frequency to decrease power-consumption and increase autonomy while on. The ADC clock at 16MHz Arduino's operating frequency is subdivided to 16MHz/128=125KHz. Furthermore, an A/D conversion requires 13 clock cycles to complete, which results to 125KHz/13=9615Hz. Consequently, the sampling rate of the proposed 6-channel EMG prototype equals to 9615/6=1602.5Hz which is higher than the range of power spectral density of the EMG signal (20Hz- 400Hz) [12]. This means that the platform can sample and acquire the EMG signal without loss even if the EMG signal had to be recorded. The restrictions of the targeted draft control applications are more loose in terms of sampling speed and resolution as can be seen by the speed of the muscle contractions displayed in Fig. 4 (each contraction lasts for more than 1 sec).

Currently, the developed prototype version simply displays the EMG acquisition results through serial port, which introduces extra delay. In the final version, higher sampling rates through network connectivity or Bluetooth will be supported.

The tasks undertaken by the Arduino uC are the following: Initially, analog to digital conversion of 6 EMG channel values takes place. The digitized EMG values of each channel (c_i) are used to update the moving average A_{ij} ($1 \le i \le 6$) using the last M values of each channel. The set of the last N, A_{ij} values ($1 \le j \le N$) spanning at a time interval of about 1 sec is examined to update the thresholds T_j that discriminate the level of contraction detected. If the trend of these N values is a rising or a falling edge, no threshold update is performed since it is assumed that a muscle contraction takes place at that time. Otherwise, these N values are expected to be around a DC offset average A_{off} that is also continuously updated. In the most robust case and for the sake of simplicity, we assume that we are only interesting in two states of each EMG channel: contraction or no_contraction. Thus, there is only one $T_j = T$ threshold that is associated with a constant offset T_{off}. The offset T is continuously updated to A_{off}-T_{off}.

The power-consumption of the 6 EMG channels is lower than 6×450 uA=2.7 mA instead of approximately 6×19.6 mA=117.6 mA that would be required if we had used all the stages displayed in Fig. 1. Arduino Zero in the tested configuration dissipates approximately 20 mA at 5 V power supply. The power consumption can be further reduced if the system enters (low-power) idle mode when sampling is not needed. Moreover, relay units could be driven by Arduino before it falls into sleep (idle) mode to turn-off the supply to the EMG channels.

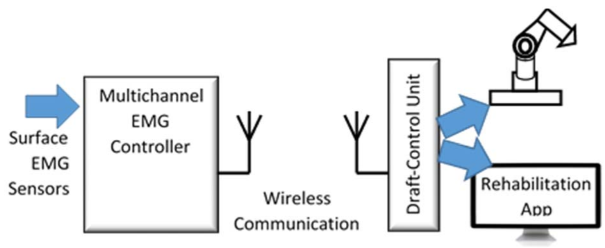

Fig. 7. The Architecture of the draft-control environment that comprises the multichannel EMG controller and the draft-control unit on the reception side.

Basic application examples of the proposed surfaced-EMG controller are shown in Fig. 7. The surface EMG sensors provide their signal values to the multichannel EMG controller. Instead of the estimation of the moving averages A_i and A_{off} described earlier, Principle Component Analysis could be additionally employed to further smooth the acquired EMG signals. EMG digital values are formatted and transmitted via wireless communication link to the receiver. On the receiver side, the draft-control unit measures user reactions and according to the received data, it can for example either move the robotic arm or instruct the patient to perform specific rehabilitation exercises. In the latter case, the display, at the receiver, coaches the patient to do exercises while the draft-controller unit rechecks the updated information sent by the EMG readout module, in response to patient reaction (rehabilitation app).

Algorithm 1: EMG values processing

```
Input:EMG1, …, EMG6 signal values
Output: Configuration code, Timestamp
Setup:
        Wireless network setup
        Timestamp = zero
        Counter=0;
Loop forever: /* Each iteration is repeated every 1/9615≈100usec */
        Retrieve ci values
        Update Aij from the last N, ci values of each channel j
        If ((Counter%1000=0) AND NoContractionNow(Aij))
            Update Aoff from the last M, Aij values of each channel j
                  /* every ≈100msec */
        T = Aoff-Toff
        time_delay
        Timestamp = time_delay
        if c1 value ≤ T then
                channel1=contraction
        else:
                channel1 = no_contraction
        …
        if c6 value ≤ T value then
                channel6=contraction
        else:
                channel6 = no_contraction
        configuration_code =lookup(channel1,…,channel6)
        transmit (Timestamp, configuration_code)
        Counter=Counter+1
End
```

Fig. 8. Pseudoalgorithm of the microcontroller operation in multichannel EMG controller.

TABLE I. CONFIGURATION CODE

Right arm	***Left arm***	***Right breast***	***Left breast***	***Right wrist***	***Left wrist***	***Code***
No contra-ction	No contra-ction	No contra-ction	No contra-ction	No contra-ction	No contra-ction	0
No contra-ction	No contra-ction	No contra-ction	No contra-ction	No contra-ction	Contra-ction	1
No contra-ction	No contra-ction	No contra-ction	No contra-ction	Contra-ction	No contra-ction	2
…	…	…	…	…	…	…
Contra-ction	Contra-ction	Contra-ction	Contra-ction	Contra-ction	Contra-ction	63

As mentioned above, high precision is not required for the draft EMG levels which in our case are used to form various control combinations. These combinations are elaborated by the draft-control unit to control e.g. a prosthetic limb or a remote robotic arm. Within this context, the digitally converted values of the EMG signals on the EMG controller can be formatted to form a configuration code as follows: If two levels of muscle status (contraction or no contraction) are recognized by each channel then we can define up to 2^6=64 instructions to drive for example the robotic arm with a corresponding number of degrees of freedom. Table I shows how the configuration code can be derived based on these 64 different combinations when the six EMG sensors are placed for instance on the arms, on the breast and on the wrists. In case that three levels of muscle status (no contraction, mild contraction, and intense contraction) can be identified at least in some EMG channels, then more combinations (up to 3^6=729) can be specified to control additional degrees of freedom in accordance to the specific requirements of the target application.

The microcontroller is programmed to process the values retrieved from the ADC converter at specific timestamps and according to a digital voltage threshold value, it identifies muscle contraction or not, as described earlier. Then, it takes into account the lookup table, which includes the information of Table I and transmits the configuration code to the draft-control unit, which decodes it and controls accordingly the rehabilitation application or the robotic arm. Fig. 8 describes the pseudoalgorithm (algorithm 1) that is executed by the microcontroller during EMG values' processing.

According to algorithm 1 in each timestamp the six EMG signals that are sensed are eventually assigned to a configuration code and transmitted as a pair (i.e. timestamp, configuration_code). The detection of the DC offset A_{off} and the threshold value T are updated in each iteration. The number M of c_i values used to estimate each A_{ij} and the number N of A_{ij} values used to estimate A_{off} are selected to be equal to 10. The update of A_{ij} and A_{off} in Algorithm 1 is implemented using a pair of small FIFO buffers (with size: M=N=10) of c_i and A_{ij} values, respectively. The sampling is performed in a uniform way, since data transmission is controlled based on the corresponding timestamp. It is denoted that transmission could also take place without assigning a specific time stamp sampling (non uniform sampling). Since draft-control applications do not require high-precision values to operate, the main advantages of our approach when compared against relevant implementations described in the literature [3, 4, 5, 6, 11] comprise ease of integration, low-cost circuitry design, low-power consumption and scalability depending on the employed number of the channels.

IV. Conclusions

Work in progress for the design of a low-cost and low-power multi-purpose surface-EMG based system for draft control applications has been presented. The proposed readout system identifies rough changes of the EMG sensors and delivers them to the microcontroller's platform. The readout system is formed by utilizing a small number of opamps in comparison to conventional designs, which translates to lower power consumption and cost. Alternatively, an INA333 instrumentation amplifier is also under consideration for using it as a first stage of the EMG readout circuit of Fig. 1, which in this case it is estimated that will reduce further the current consumption. The microcontroller processes the converted by the ADC data and transmits it to the draft-controller unit that instructs the target application. Future work involves the implementation of the full read-out circuit, its measurements and interfacing to a microcontroller board. Moreover, the implementation of the draft-control receiver is also to take place together with the connection to a rehabilitation application. Finally, our target is to apply the proposed system to patient rehabilitation and perform a clinical study based on this system.

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